

³⁰
Question 3: [40 points]

Billionaire Bob gets utility from two activities: playing golf (G) and traveling to exotic places with his private jet (J). Bob's preferences are represented by the utility function:

$$U(G, J) = G^{0.5} J^{0.5}$$

where G is the number of rounds of golf per year, and J is the number of private jet trips per year. Bob has \$120,000 to spend on these two activities every year. The price of one round of golf (P_G) is \$200 and the price of one private jet trip (P_J) is \$4,000.

- ✓ a) Find Bob's utility-maximizing consumption bundle of golf and private jet travel. Also illustrate the solution in a graph.

- ✓ b) Suppose that the market demand for private jet travel (for Bob and similar high-income individuals) is given by:

$$P = 10,000 - 500Q$$

where P now is the price per private jet trip (in \$) and Q is the number of private jet trips per year (in thousands). The marginal cost of providing a private jet trip is \$4,000. In addition, each private jet trip creates CO₂ emissions that impose a marginal external cost (MEC) to society equal to \$2,000. However, this is not a cost that Bob (or his rich buddies) bother about.

How many jet trips will be made in total? How large is consumer, producer and total surplus in society? ^{thousands}

- ✓ c) Suppose the government asks you for advice on how to reduce CO₂ emissions from private jet trips. Please explain the recommendation you would provide to the government, and how your advice would ensure that the socially optimal number of private jet trips is reached. Illustrate your answer in a graph.
- ✓ d) How would your advice affect Bob individually? In particular, how would it affect his optimal allocation of golf and private jet trips. How would it affect his utility? Illustrate in a graph or provide the exact solution using math.

Question 1. [40 points]

A hairdresser operates as a local monopolist in a small town. The hairdresser serves two distinct groups of customers: men and women, who have different demand for haircuts.

The demand functions are given by:

$$\text{Men: } Q_M = 100 - 2P_M$$

$$\text{Women: } Q_W = 120 - P_W$$

where Q is the number of haircuts per week and P is the price per haircut (in NOK).

The marginal cost of providing a haircut is constant and equal to 20 NOK for both men and women.

- ✓ a) Assume the hairdresser can perfectly segment the market and charge different prices to men and women. Derive the profit-maximizing price and quantity for each group. Also calculate the producer and consumer surplus, for each customer segment and in total. ✓
- ✓ b) Suppose the hairdresser is not allowed to price discriminate and must charge a single price to all customers. Derive the total market demand and find the profit-maximizing price and quantity. Also calculate producer surplus and consumer surplus under uniform pricing.
- ✓ c) Compare the outcomes under price discrimination and uniform pricing. Please discuss the effect on prices, quantity, producer and consumer surplus. Who benefits from price discrimination? Is society better off with unit pricing?

Suppose that anti-discrimination rules make it illegal for the hairdresser to charge different prices to men and women. In response, the hairdresser decides to offer two versions of the service instead:

- Standard haircut (S): haircut only
- Premium treatment (T): haircut + wash + styling

Assume each customer buys at most one service. We also assume that a representative man is willing to pay 50 NOK for a standard haircut and 60 NOK for a premium treatment. A representative woman, on the other hand, is willing to pay 70 NOK for a standard haircut and 90 NOK for a premium treatment.

The marginal cost for the standard haircut is still 20 NOK, but the extra effort increases the marginal cost of the premium treatment to 30 NOK.

- ✓d) Suppose the hairdresser sets the following prices for the two versions:

$$P_S = 35$$

$$P_T = 70$$

Will men and women self-select into the intended services, where men buy the standard haircut (S) and women buy the premium treatment (T)? Explain.

- ✓e) Which are the highest prices the hairdresser can set while at the same time ensuring that men buy the standard haircut and women the premium treatment? Given these prices, which group is happiest with the anti-discrimination policy: Men, women or the hairdresser?

Question 2. [30 points]

Two firms, Jarlsberg and Norvegia, produce and sell differentiated types of cheese in the same domestic market (i.e. Norway). Due to slight differences in taste and branding the products are not perfect substitutes. The demand for each firm (J =Jarlsberg; N =Norvegia) depends on both its own price and the price of the competing firm:

$$Q_J = 100 - P_J + 0.5P_N$$

$$Q_N = 100 - P_N + 0.5P_J$$

where Q_J is the quantity sold by Jarlsberg, and P_J is the price for Jarlsberg cheese (whereas Q_N and P_N denote the quantity and price for Norvegia).

Both firms have the same constant marginal cost, which we for simplicity can assume is zero. We also assume that both firms simultaneously choose their prices.

- ✓a) Derive the profit-maximizing price for each firm, and calculate quantities sold and profits.

Suppose Jarlsberg launches a successful advertising campaign, which results in the following new demand function for Jarlsberg cheese:

$$Q_J = 100 - 0.5P_J + 0.5P_N$$

while Norvegia's demand remains unchanged

- ✓b) Derive the new equilibrium prices, quantities and profits.
- ✓c) Compare to your answer in a) and explain the intuition for what has happened.
- ✓d) Norway has for a long time had large import tariffs on cheese (up to 277% or more). After negotiations with the European Union, Norway's government decides to remove its cheese tariffs. Explain intuitively how this decision will affect the cheese market in Norway. In particular, what kind of competition is likely to emerge, and how will that affect the demand and pricing decision for Jarlsberg and Norvegia?